Inner Product

* Inner Product Over Field $\mathbb{R}$:

Let, V be a vector space over the field $\mathbb{R}$.
An inner product on V is a function that takes each ordered pair $(\vec{u}, \vec{v})$ of elements of V to a number $\langle \vec{u}, \vec{v} \rangle \in \mathbb{R}$ and has the following properties:

i) positivity: $\langle \vec{u}, \vec{u} \rangle \geq 0$ for all $\vec{u} \in V$.

ii) definiteness: $\langle \vec{u}, \vec{u} \rangle = 0$ if and only if $\vec{u} = 0$.

iii) additivity in first slot:
$\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ for all $\vec{u}, \vec{v}, \vec{w} \in V$

iv) homogeneity in first slot:
$\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$ for all $\lambda \in \mathbb{R}$ and for all $\vec{u}, \vec{v} \in V$.

v) Symmetry: $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$ for all $\vec{u}, \vec{v} \in V$.

* Inner Product over the field $\mathbb{C}$:

Let, V be a vector space over the field $\mathbb{C}$.
An inner product on V is a function that takes each ordered pair $(\vec{u}, \vec{v})$ of elements of V to a number $\langle \vec{u}, \vec{v} \rangle \in \mathbb{C}$ and has the following properties:

i) Positivity: $\langle u, u \rangle \geq 0$ for all $\vec{u} \in V$.

ii) Definiteness: $\langle \vec{u}, \vec{u} \rangle = 0$ if and only if $\vec{u} = 0$.

iii) Additivity in first slot:

$\langle \vec{u}+\vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ for all $\vec{u}, \vec{v}, \vec{w} \in V$.

iv) Homogeniety in first slot:

$\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$ for all $\lambda \in \mathbb{C}$ and for all $\vec{u}, \vec{v} \in V$.

v) Conjugate Symmetry: $\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$ for all $\vec{u}, \vec{v} \in V$.

* Inner Product in $V = \mathbb{R}^n$:

* Standard Inner Product in $\mathbb{R}^2$:

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = x_1 y_1 + x_2 y_2 \quad \text{[Dot Product]}$$

$$\left\langle \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 7 \end{pmatrix} \right\rangle = (2 \times 4) + (3 \times 7) = 8 + 21 = 29.$$

* Standard Inner Product in $\mathbb{R}^n$:

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right\rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = \sum_{i=1}^{n} x_i y_i$$

Q. Show that $\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 2x_1 y_1 + 3x_2 y_2$ is an inner product in $\mathbb{R}^2$. [Yes].

Q. Show that $\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 2x_1 y_1 - 3x_2 y_2$ is not an inner product in $\mathbb{R}^2$. [No]

Ans: $\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\rangle = 2.1.1 - 3.2.2 = 2 - 12 = -10 \neq 0$

* Inner Product $V = C^n$.

* Standard Inner Product in C^2 :-

$$\left\langle \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \right\rangle = z_1\bar{w}_1 + z_2\bar{w}_2$$

$$\left\langle \begin{pmatrix} 2+i \\ 7 \end{pmatrix}, \begin{pmatrix} 2+i \\ 7 \end{pmatrix} \right\rangle = (2+i)\overline{(2+i)} + 7.\bar{7} = 4+1+49 = 54 > 0.$$

* Standard Inner Product in C^n.

$$\left\langle \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \right\rangle = z_1\bar{w}_1 + z_2\bar{w}_2 + \cdots + z_n\bar{w}_n .$$

* Inner Product in Matrix Spaces:

* Standard Inner Product of 2X3 Real Matrices:

$$\langle A, B \rangle = \text{Trace}(A^T B).$$

$$\left\langle \begin{pmatrix} 4 & 5 & 2 \\ 1 & 0 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 & 5 \\ 4 & 2 & 3 \end{pmatrix} \right\rangle$$

$$= \text{Trace}\left[\begin{pmatrix} 4 & 1 \\ 5 & 0 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 & 5 \\ 4 & 2 & 3 \end{pmatrix} \right]$$

$$= \text{Trace} \begin{pmatrix} 12 & 6 & 23 \\ 10 & 5 & 25 \\ 16 & 8 & 19 \end{pmatrix}$$

$$= 12 + 5 + 19 = 36 .$$

* Standard Inner Product of $m \times n$ Real Matrices

$$\langle A, B \rangle = \text{Trace}(A^T . B)$$

* Inner Product in Function Spaces:

* Inner Product in the space of integrable functions:

* $\langle f(x), g(x) \rangle = \int_a^b f(x) . g(x)\, dx$ [Real]

* $\langle f(z), g(z) \rangle = \int_a^b f(z) . \overline{g(z)}\, dz$ [complex]

Q. $\langle f(x), g(x) \rangle = \int_0^1 f(x) . g(x)\, dx$; Find the inner product of e^x and x.

Ans: $\langle e^x, x \rangle = \int_0^1 e^x . x\, dx = \left[x e^x - e^x \right]_0^1$

$= e^1 - e^1 - (-e) = e$.

* Norm (Abstraction of length):

Let, V be a vector space over the field F.
A Norm V is a function that takes each $\vec{u}$ of element of V to a number $\|\vec{u}\| \in \mathbb{R}$. and has the following properties:

i) Positivity: $\|\vec{u}\| \geq 0$ for all $\vec{u} \in V$.

ii) Definiteness: $\|\vec{u}\| = 0$ if and only if $\vec{u} = 0$.

iii) Triangle Inequality: $\|\vec{u}+\vec{v}\| \leq \|\vec{u}\|+\|\vec{v}\|$ for all $\vec{u},\vec{v} \in V$.

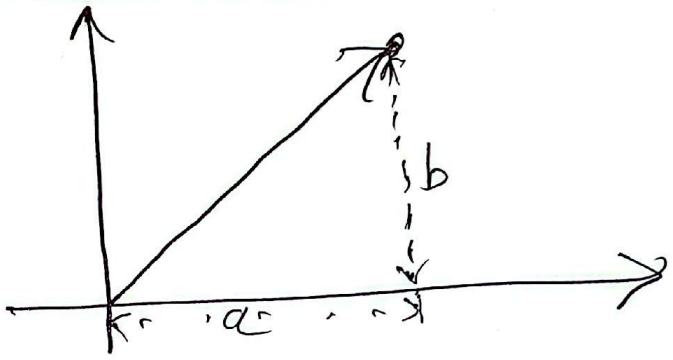
iv) Homogeniety: $\|\lambda\vec{u}\| = |\lambda|\|\vec{u}\|$ for all $\lambda \in F$ and for all $\vec{u} \in V$.

* Length from Inner Product:

$$\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}.$$

* Standard Euclidean Norm in $\mathbb{R}^2$:

$$\vec{u} = \begin{pmatrix} a \\ b \end{pmatrix}.$$



$$\|\vec{u}\| = \sqrt{a^2+b^2} = \sqrt{\langle \vec{u}, \vec{u} \rangle}$$

$$\langle \vec{u}, \vec{u} \rangle = a^2+b^2$$

* Standard Euclidean Norm in $\mathbb{R}^n$:

$$\vec{u} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad \|\vec{u}\| = \sqrt{a^2+b^2+c^2}$$

$$\vec{u} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \qquad \|\vec{u}\| = \sqrt{x_1^2+x_2^2+\dots+x_n^2}$$

$$\|\sin(x)\| = \sqrt{\langle \sin(x), \sin(x) \rangle} = \sqrt{\int_{-\pi}^{\pi} \sin(x)\cdot\sin(x)\,dx}$$

Orthogonality

Let, V be a vector space over the field F and $\langle \vec{u}, \vec{v} \rangle$ be an inner product. Two vectors $\vec{u}$ and $\vec{v}$ are called orthogonal if:

$$\boxed{\langle \vec{u}, \vec{v} \rangle = 0}$$

Orthogonality of Vectors

* Orthogonal Vectors in $\mathbb{R}^n$:

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right\rangle = x_1y_1 + x_2y_2 + \cdots + x_ny_n = 0$$

$$= \sum_{i=1}^{n} x_i y_i = 0 \quad \text{[For Orthogonality]}$$

Q. * Show that the vectors $(1,-1,0), (1,1,-2)$ are orthogonal with respect to standard inner product in $\mathbb{R}^3$,

Sol^n: $\langle (1,-1,0), (1,1,-2) \rangle = 1.1 + (-1).1 + 0(-2) = 0$

$\therefore (1,-1,0)$ and $(1,1,-2)$ are orthogonal.

Q. Show that $(2+i, 1-3i)$ and $(10, 1+7i)$ are orthogonal with respect to standard inner product in $\mathbb{C}^2$.

Sol^n $\langle (2+i, 1-3i), (10, 1+7i) \rangle = (2+i).\overline{10} + (1-3i).\overline{(1+7i)}$

$= (2+i).10 + (1-3i)(1-7i)$ | $\overline{a+ib} = a-ib$

$= \cancel{(2+i).10}\ 20+10i+1-7i-3i-21$

$= 0$

$\therefore (2+i, 1-3i)$ and $(10, 1+7i)$ are orthogonal

Orthogonality of Functions

$\langle f(x), g(x) \rangle = \int_a^b f(x).g(x)\,dx = 0$ [For Orthogonality]

Q. Consider the inner product:

$\langle f(x), g(x) \rangle = \int_{-1}^{1} f(x)\,g(x)\,dx$

Show that the functions $3x^2-1$ and $5x^3-3x$ are orthogonal.

Soln: $\langle 3x^2-1, 5x^3-3x \rangle = \int_{-1}^{1} (3x^2-1)(5x^3-3x)\,dx$

$= \int_{-1}^{1} (15x^5 - 9x^3 - 5x^3 + 3x)\,dx$

$= \left[\frac{15x^6}{6} - \frac{9x^4}{4} - \frac{5x^4}{4} + \frac{3x^2}{2}\right]_{-1}^{1}$

$= \left(\frac{15}{6} - \frac{9}{4} - \frac{5}{4} + \frac{3}{2}\right) - \left(\frac{15}{6} - \frac{9}{4} + \frac{5}{4} + \frac{3}{2}\right)$

$= 0$ $\therefore$ functions are orthogonal.

Q. Consider the inner product

$$\langle f(x), g(x) \rangle = \int_{-\pi}^{\pi} f(x) g(x)\, dx .$$

Show that the functions $\sin(x)$ and $\cos(2x)$ are orthogonal.

Soln: $\langle \sin(x), \cos(2x) \rangle = \int_{-\pi}^{\pi} \sin(x) \cdot \cos(2x)\, dx$

$$= \frac{1}{2} \int_{-\pi}^{\pi} 2 \sin(x) \cos(2x)\, dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \left[\sin(x+2x) + \sin(x-2x) \right] dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \sin(3x)\, dx + \frac{1}{2} \int_{-\pi}^{\pi} \sin(-x)\, dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \sin(3x)\, dx - \frac{1}{2} \int_{-\pi}^{\pi} \sin(x)\, dx$$

$$= \frac{1}{2} \left[-\frac{\cos(3x)}{3} \right]_{-\pi}^{\pi} - \frac{1}{2} \left[-\cos(x) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2} \left[\frac{-\cos(3\pi)}{3} - \frac{-\cos(-3\pi)}{3} \right] - \frac{1}{2} \left[(-\cos\pi) - (-\cos(-\pi)) \right]$$

$$= \frac{1}{2} \left[\frac{-(-1)}{3} - \frac{-(-1)}{3} \right] - \frac{1}{2} \left[(-1) - (-1)(-1) \right]$$

$$= 0$$

$\therefore$ functions are orthogonal.